

Rotary Brush Tool Effectively Filters SLF from Grape Stream

Team DEAd_heAD

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Problem Overview

During short, high-volume harvest periods, grape farmers must remove invasive spotted lantern flies (SLFs) from grape vines quickly, yet current methods continue to miss insects. The challenge is to design a separation system that can reliably distinguish SLF from grapes within a mixed and variable product stream.



Figure 1. Infestation of lanternflies on grape vine. The invasive species flock to grapes during harvest season to feed.

There are three opportunities to remove SLF: before, during, and after harvest processing. Our approach is implemented during the harvesting stage, when the grapes pass from the harvester on a conveyor belt. By looking for solutions to apply at this stage, we reduce the number of devices needed to effectively remove SLF. However, harvested grapes enter the conveyor in different forms, including clusters, loose berries, as well as leaves, stems, soil, and other debris, making accurate identification and removal complex.



Figure 2. Conveyor belt processing grapes during harvesting. Large bunches of grapes gather together on the belt.

Our system must operate effectively within this dynamic flow while maintaining high throughput and compatibility with existing processing steps. Possible risks to consider are product loss (excessive grape removal, damage, or unwanted contamination), increased processing time (slowing down throughput with extra processes/systems), and clogging/equipment failure (grapes, insects, or debris obstructing our machine). To mitigate these risks, we focused on a rotary brush tool that is mechanically simple, cost-effective, and fits seamlessly into current harvesting lines.

Success Criteria

Our design was a rotary brush tool that can be attached to current harvester conveyor belts to remove SLF from grapes just harvested from vines. The criteria we sought out was:

- 1) The brushes should not damage, crush, or visibly harm more than 5% of grapes. Our clients have emphasized product loss as a major problem in current SLF removal systems. We factored this in by simulating real grapes through our prototype and showing the number of grapes harmed vs. the total number of grapes passed through our prototype.
- 2) The brushes should be tuned to remove at least 90% of SLF from grapes in a single pass. High removal rates are a high priority to prevent product rejection.
- 3) The system should be able to withstand realistic operation without shifting more than 5mm. Greater stability is a medium priority.
- 4) The system should be able to process grapes on a conveyor belt at a speed of 1 m/s. Higher speeds are a medium priority.

Testing Results for Initial Prototype

Our initial prototype consisted of food grade brushes rotating on a wooden dowel that is driven by a standard hand held drill. It is mounted above a conveyor system to simulate harvesting conditions. The main mechanical risks involve brush motion and structural stability. The following tests allowed us to identify three factors which strongly impacted performance: brush length, rotation speed, and rotation direction.



Figure 3. Initial Prototype of Mount and Brush Mechanism. The wooden dowel is driven by a handheld drill.

During testing we made a mock conveyor belt and modeled grapes out of clay, each of the approximate weight of a grape (5-7g). We modeled SLF with loops of masking tape, approximating their weight and clinging tendency. Using the cordless drill to drive the brushes, we measured how many SLF were knocked off and how much our mounting moved. By changing the direction of the drill, we also tested whether a clockwise or counterclockwise brush rotation was more effective for SLF removal.



Figure 4. Initial Prototype Conveyor and Grapes. The grapes are modeled with clay and the SLF as masking tape on top of a mock cardboard conveyor.

Drill Spec Test - 3" Brush		
Direction \ Speed	500 RPM	2000 RPM
Clockwise	3/8	6/8
Counterclockwise	0/8	0/8

Table 1. Speed and direction of brush rotation greatly impact effectiveness. 2000 RPM was the most successful speed.

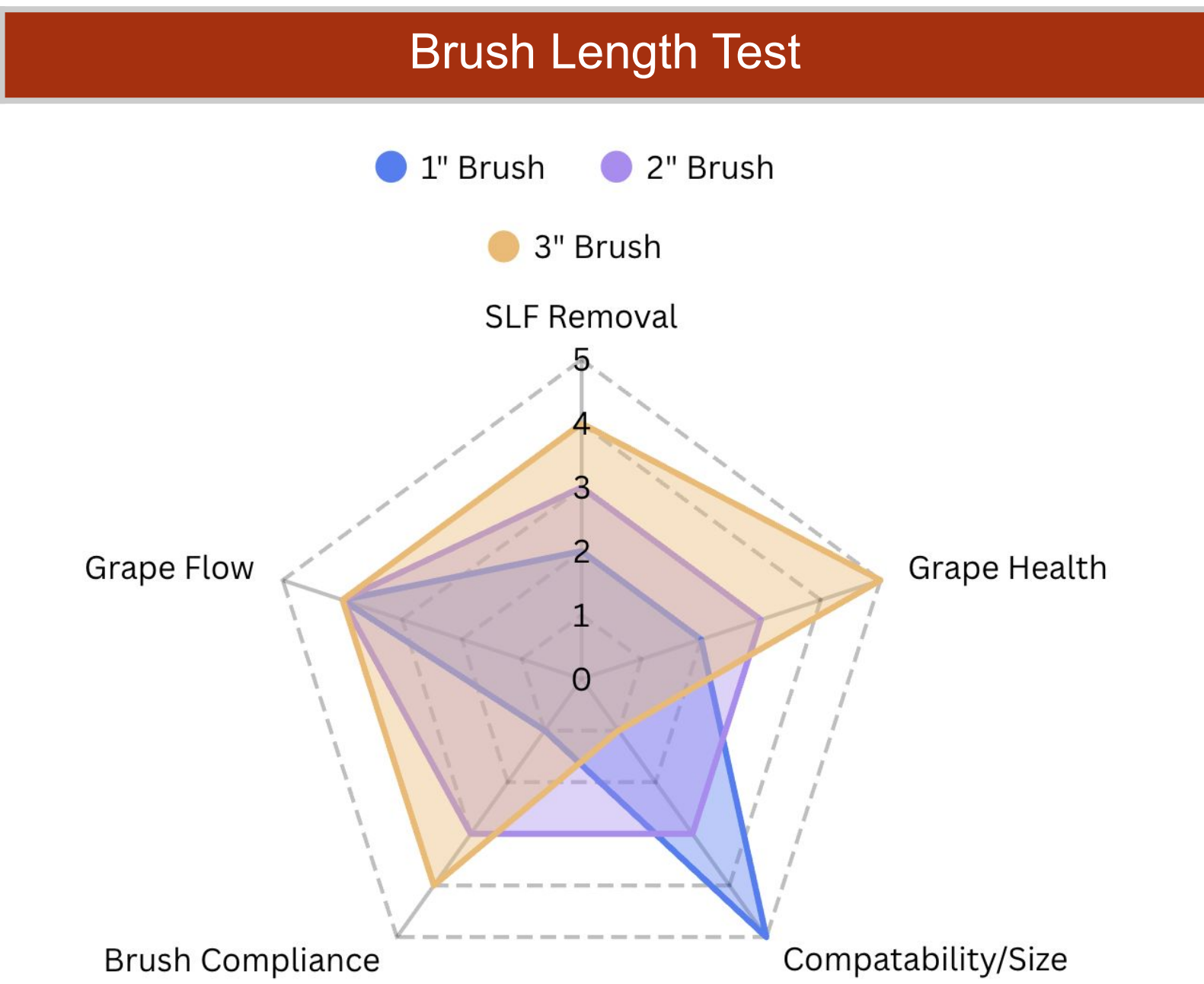


Table 2. Qualitative results of Brush Length Test. The 3" brush is most well-rounded choice.

Our testing highlighted several areas in need of improvement. For drill speeds >500 RPM, our prototype started shifting, indicating a need for a sturdier, more robust mounting design. As we deduced that 2000 RPM in the clockwise direction removed the most SLF, we needed sturdier mounts on future prototypes. The 3" brush was identified as the most effective choice, as shown in Table 2, which should also be integrated in future configurations.

Final Prototype

The initial prototype established baseline estimates for dimensions, operating speed, and overall feasibility. The final prototype builds on this by closely replicating a real harvester conveyor system with an integrated rotary brush mechanism. A mock conveyor section was constructed with dimensions similar to an actual harvesting belt, using food-grade materials for the belt and brushes, along with 3D-printed components. The mounts are machined from aluminum, enabling precise placement within a 0.010-inch tolerance and ensuring structural stability. The system is driven by a 3000 RPM motor paired with a speed controller, allowing fine adjustment of operating speed to meet specific testing requirements.

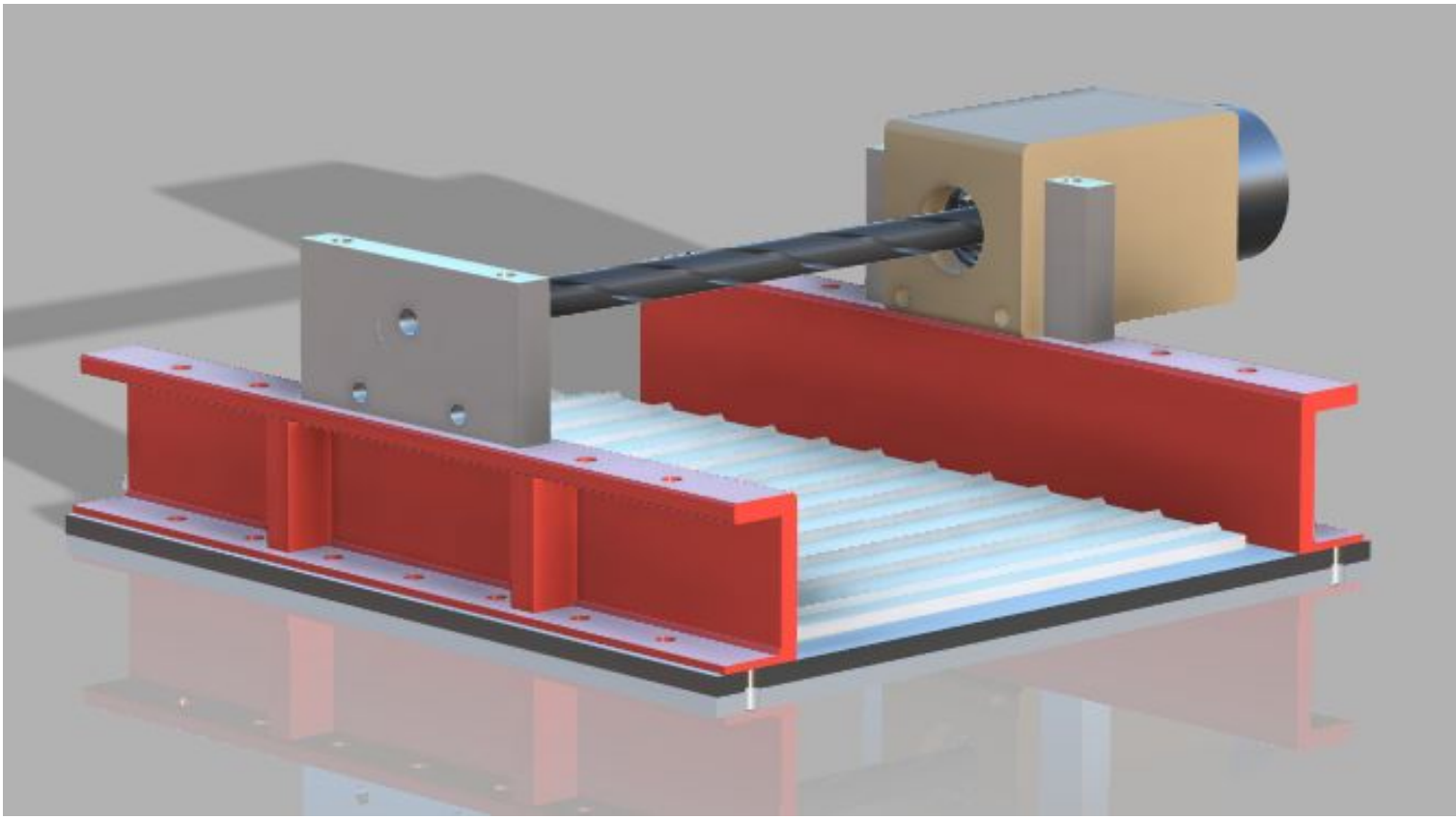


Figure 5. CAD Render of Final Prototype. Pictured is the improved walls, conveyor belt, and rod.

Additionally, the final prototype incorporates a shaft with an internal helical channel for mounting the brushes. This design increases stability by maximizing the brush mounting surface area. In contrast, the first prototype used four externally attached rod supports (shown in Figure 3), whereas the final design provides a continuous mounting interface.

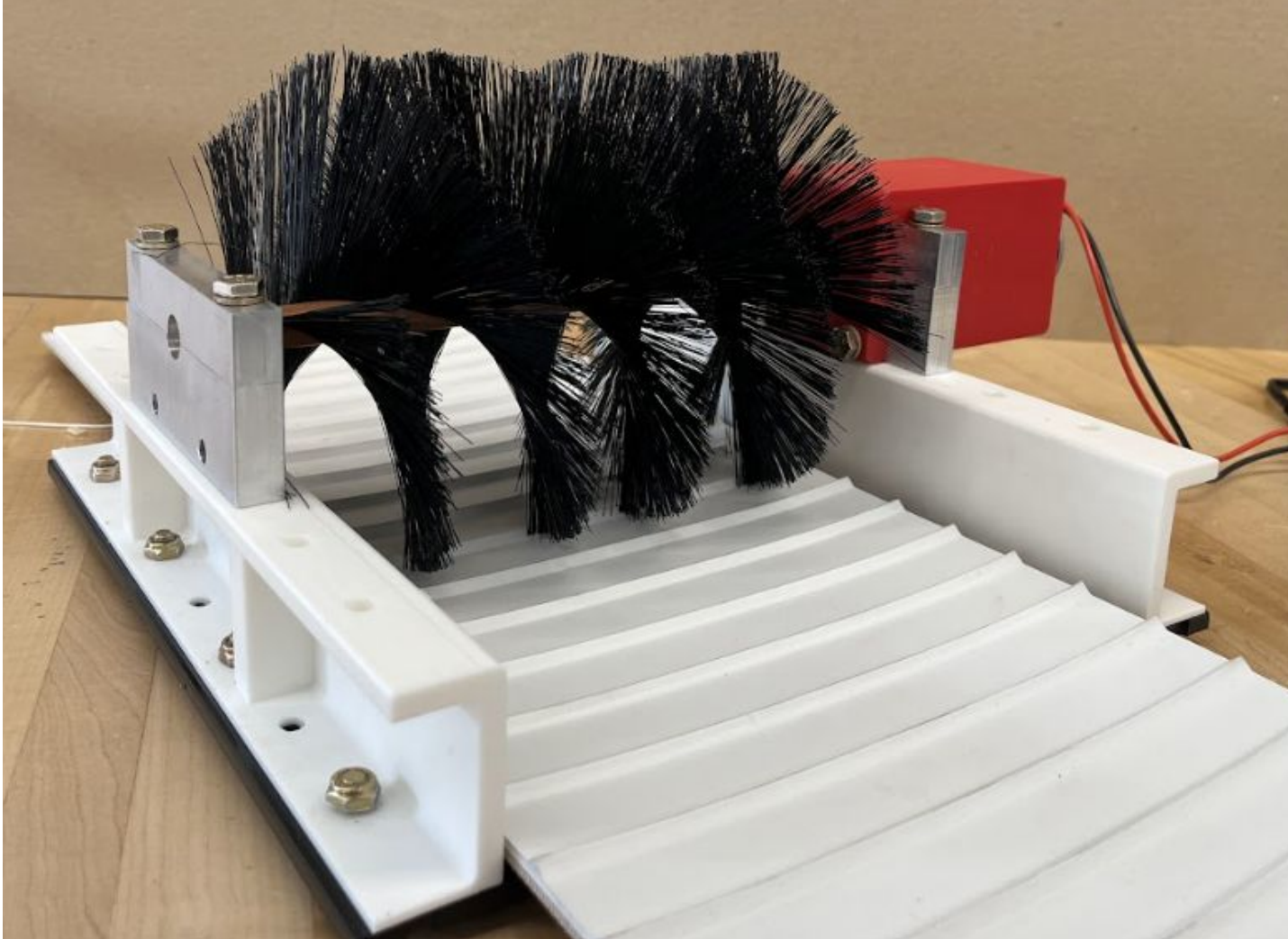


Figure 6. Final Prototype Assembly. The mounts are aluminum while the rod and motor casing are 3D printed.

Scan Below for a Demonstration



The assembly and use of the final prototype demonstrated several key advantages: The setup is intuitive, requires minimal steps, and maintains precise alignment. Additionally, bolt access is straightforward, making fastening, disassembly, and maintenance simple and efficient. In operation, the final prototype showed improved stability over the initial design. At speeds above 500 RPM, there was no noticeable vibration and the helical shaft design kept the brush mounts secure without shaking. Overall, the final prototype demonstrated significant improved rigidity and operational consistency than the initial prototype.

Conclusion

We researched the already existing harvesters and found the OXBO 6030 was already equipped with PREMIUMDESTEM and PREMIUMSORT technology that included a majority of what our product was trying to achieve. Therefore, as our results prove that the brush mechanisms are indeed brushing off SLF with increased success and efficiency, further improvements can be added to the already established technology.



Figure 7. Components of existing debris removal technology. (Image found from OXBO.com about 6030)

Solution Direction and Recommendations

Future iterations should explore incorporating multiple brush stages with varying stiffness or rotational speeds to improve removal efficiency while minimizing damage to the grapes. Introducing a staged approach would allow for more controlled, gradual removal of SLF across the process. A practical implementation should include an integrated cleaning mechanism to prevent debris buildup and brush clogging, ensuring consistent performance and reducing maintenance requirements over time. Software-based control can also improve performance significantly; for example, a closed-loop feedback system, using vision-based sensors, could enable real-time adjustment brush speeds/torque based on SLF presence.

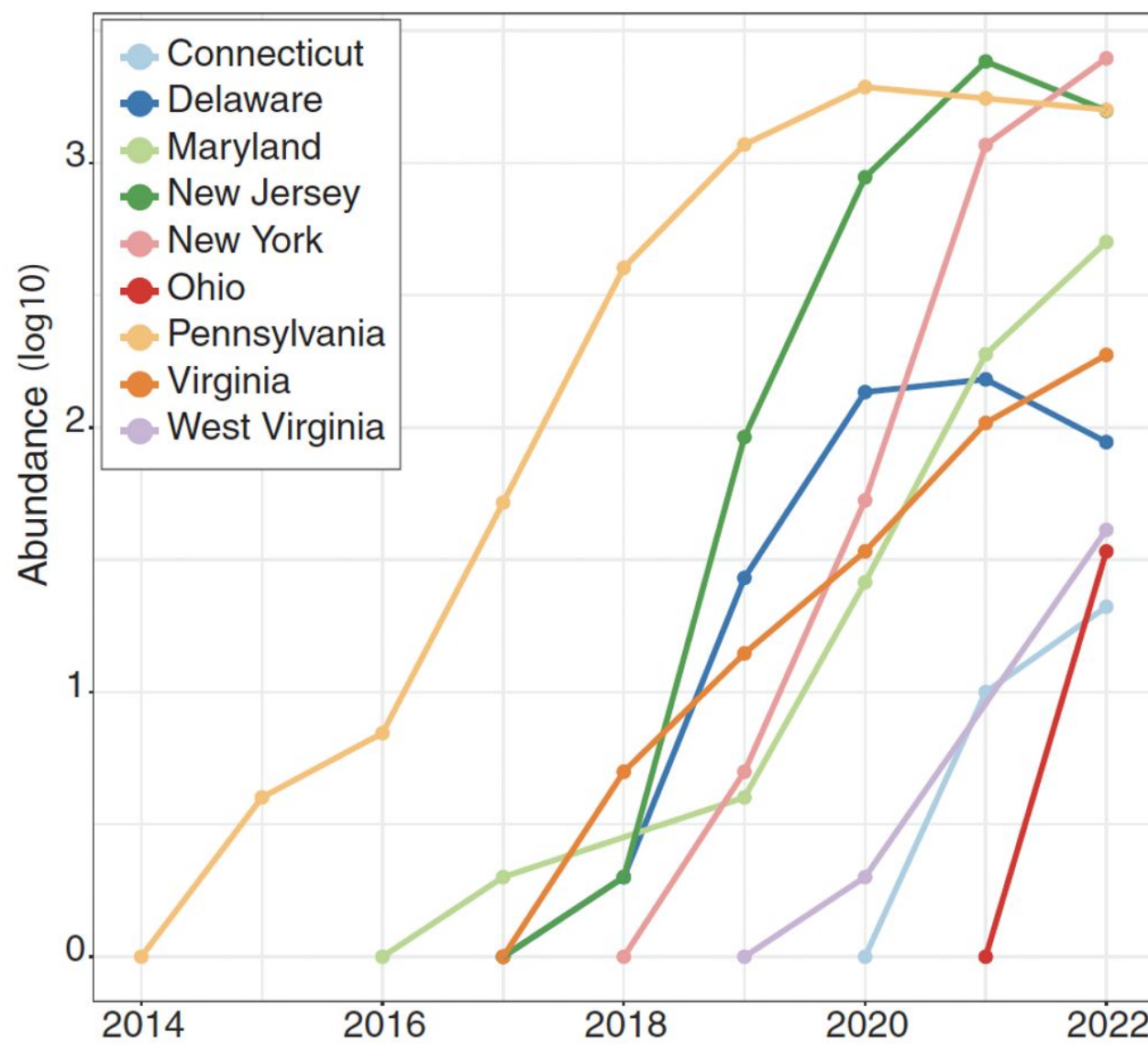


Figure 8. Graph shows increasing SLF population. (Image appears in Biological Journal of the Linnean Society.)

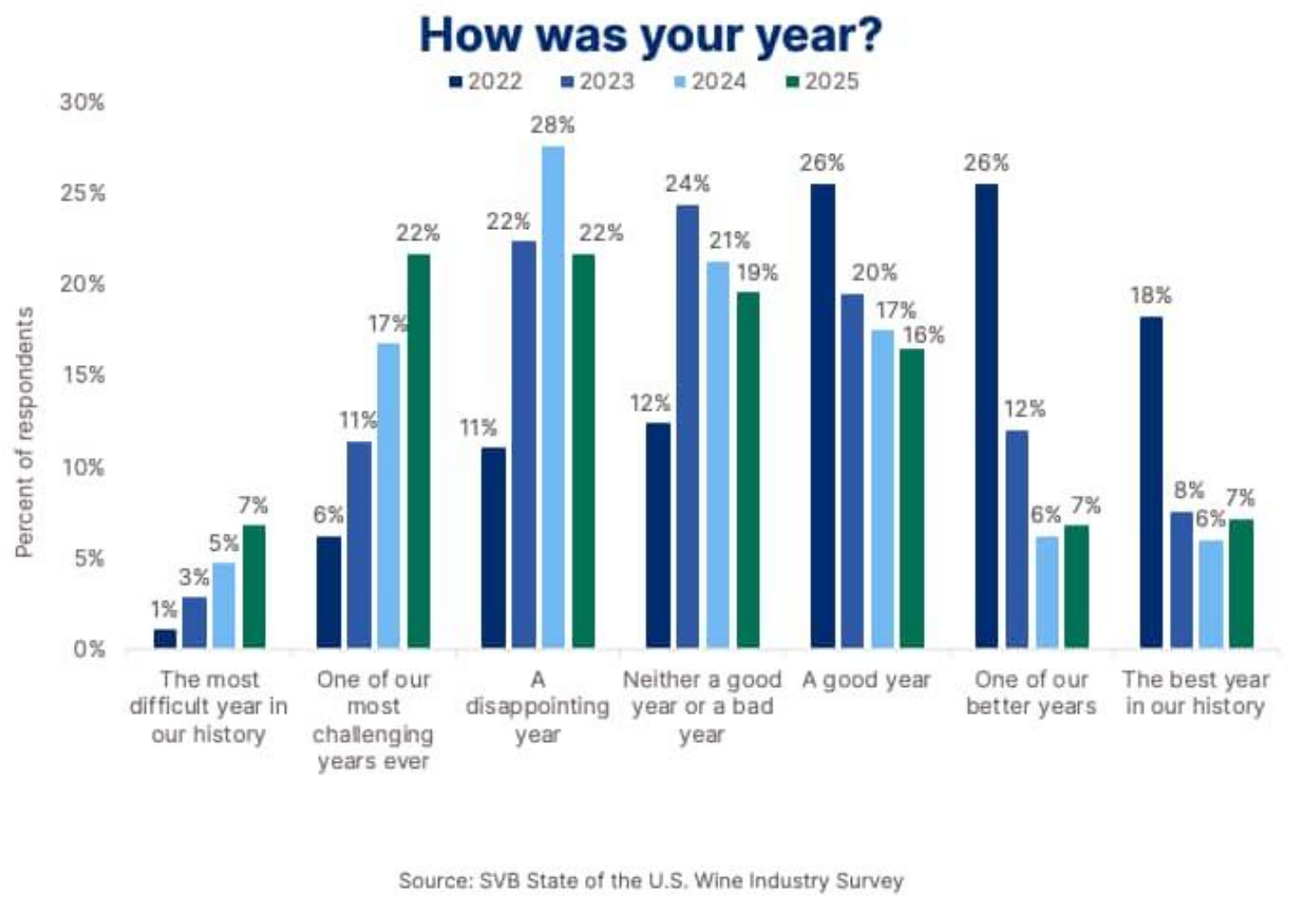


Figure 9. Chart of US Wine Industry satisfaction. (From State of the US Wine Industry Report 2026.)

The wine industry is suffering due to SLF contaminating grapes, and our prototype offers a promising path toward an effective, scalable solution.